

Lab: Mapping the Organizational Markov Blanket

Objectives

Learning Goal: Apply the Markov blanket framework to a real organization, identifying boundaries, information flows, and nested systems.

This is a **case study exercise** designed for business professionals. No coding is required — the focus is on analytical mapping and strategic insight.

Part 1: Select Your Organization (10 min)

Choose an organization to analyze. This can be:

- Your own company or department
- A well-known public company (e.g., Amazon, Toyota, a local hospital)
- A nonprofit organization you are familiar with

Tip: If you choose your own organization, you will generate the most actionable insights.

Write your response here

Part 2: Map the Markov Blanket (20 min)

Using the framework from the module, identify and document the following for your chosen organization:

2A: Sensory States — What Comes In

List 8–10 information flows that enter the organization from the external environment.

#	Sensory State	Source	Frequency	Quality Assessment
1				
2				
3				
4				
5				

Write your response here

2B: Active States — What Goes Out

List 8–10 actions, outputs, or communications that the organization directs at its environment.

#	Active State	Target	Frequency	Strategic Purpose
1				
2				
3				
4				
5				

Write your response here

2C: Internal States — What Stays Inside

List 5–7 internal states that constitute the organization's identity, knowledge, and capabilities.

Write your response here

Part 3: Assess Boundary Health (15 min)

Now evaluate the health of your organization's boundary:

3A: Boundary Rigidity Assessment

Question	Answer
Are there important external signals the organization is failing to sense ?	
Are there areas where the boundary is too porous (leaking proprietary info, letting in noise)?	
Are there internal states that should be externalized (shared with partners, customers)?	
Are there external states the org treats as internal (false assumptions of control)?	

Write your response here

3B: Nested Systems Map

Sketch the nested system structure (you can use text, bullet points, or draw and describe):

```
Enterprise Level
├── [Division/Department A]
│   ├── [Team A1]
│   └── [Team A2]
├── [Division/Department B]
│   ├── [Team B1]
│   └── [Team B2]
└── ...
```

For two of these nested systems, identify:

- What is a "sensory state" for the team but an "active state" for the individual?
- What is a "sensory state" for the division but an "active state" for the team?

Write your response here

Part 4: Homeostatic Variables (10 min)

Identify 4–6 critical variables that your organization must keep within viable bounds:

Variable	Current State	Lower Danger Zone	Upper Danger Zone	Monitoring Mechanism

Write your response here

Part 5: Synthesis and Recommendations (15 min)

Based on your analysis, answer:

1. **Biggest blind spot:** What is the most important signal your organization is currently failing to sense or underweighting?

Write your response here

1. **Boundary redesign:** If you could change one aspect of your organization's Markov blanket (add a new sensory channel, remove a barrier, strengthen a boundary), what would it be and why?

Write your response here

1. **Nested system friction:** Where is the biggest information bottleneck in your organization's nested structure? What organizational change would reduce it?

Write your response here

Summary

Analysis Component	Key Finding
Strongest boundary feature	
Biggest boundary vulnerability	
Most critical homeostatic variable at risk	
Recommended boundary redesign	

Write your response here

Debrief Questions

- How did the Markov blanket framework change how you think about your organization's boundaries?
- Were there aspects of your organization's boundary that surprised you when laid out explicitly?
- How could this framework inform a strategic planning session at your organization?